Plaid v1 — Cut Plan

Two end-grain plaid cutting boards built from cherry, walnut, and brown maple.

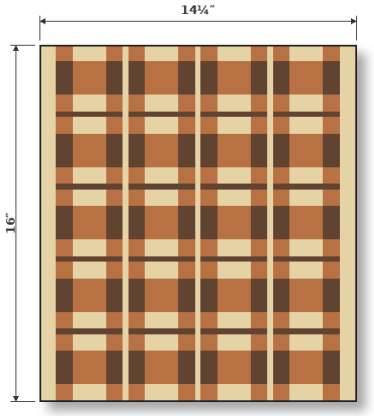
Cut list for the finished design, in build order.

1. Warnings

#	Lvl	Type	Message
1	ERROR	Crosscut Overflow	Board BC1 is over-cut by 1⅞″: its blanks demand more length than the raw board has.
2	ERROR	Crosscut Overflow	Board BC2 is over-cut by 1⅞″: its blanks demand more length than the raw board has.
3	ERROR	Over Rip	Cut list for board BM1A over-rips by 2⅛″: the strips are wider than the board.
4	ERROR	Over Rip	Cut list for board BM1B2 over-rips by 1³⁄₁₆″: the strips are wider than the board.
5	WARNING	Ghost Boards	1 board(s) must be bought (not in inventory): GH-C.
6	WARNING	Small Margin	Board BC2A leaves only ⅛″ margin (< ¼″) — too tight; prone to under-supplying strips if the board runs narrow.
7	WARNING	Zero Offcut	41 strip(s) crosscut to an exact fit with 0″ offcut — no margin for cutting error in the crosscut phase: BM1A ×6, BM1B ×3, BM1B1 ×3, BM1B2 ×10, BC1A ×5, BC2A ×6, BW1A ×3, BW2A ×5. A cut even slightly long, or kerf drift, will make the last piece short. Consider longer blanks or fewer pieces per strip.
8	WARNING	Planer Gang Too Small	There aren't enough 3″ strips to gang together to safely pass through the planer without tipping over. You might need sacrificial wood to tape onto it in order to prevent movement.

2. Finished board pattern

2 finished boards. Final thickness 1¼″.



3. Volume Sanity Check

Review the per-species volumes against inventory.

Species	Available	Required	Volume Offset
maple	5.6 BF	1.4 BF + 1.0 BF loss = 2.4 BF	+3.1 BF (229%)

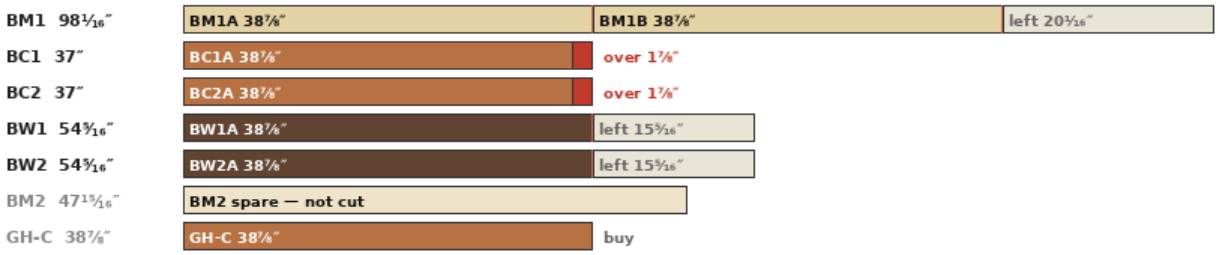
Species	Available	Required	Volume Offset
cherry	2.9 BF	1.6 BF + 1.0 BF loss = 2.6 BF	+0.3 BF (111%)
walnut	3.8 BF	1.0 BF + 0.7 BF loss = 1.7 BF	+2.0 BF (218%)

4. Crosscut Blanks

This plan involves cutting blanks from the inventory — review the inventory and the planned cuts.

Raw board	Species	Raw length	Blanks (id = length)	Leftover
BM1	maple	98 1/16″	BM1A = 38 7/8″; BM1B = 38 7/8″	20 1/16″
BC1	cherry	37″	BC1A = 38 7/8″	-1 7/8″ (overflow!)
BC2	cherry	37″	BC2A = 38 7/8″	-1 7/8″ (overflow!)
BW1	walnut	54 5/16″	BW1A = 38 7/8″	15 5/16″
BW2	walnut	54 5/16″	BW2A = 38 7/8″	15 5/16″
BM2	maple	47 15/16″	— (untouched spare)	—

Crosscut Diagram



5. Initial board rips (the cut list)

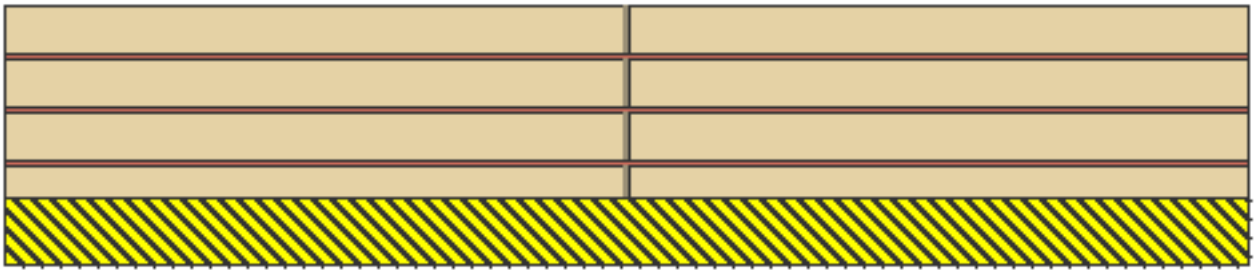
Review the rip list.

Board	Origin	Species	Length	Rips (width × count)	Used	Loss	Margin
BM1A @ 6″	BM1	maple	38 7/8″	1½″ ×4 · ¾″ ×1 · ¼″ ×1	7″	1⅛″	-2⅛″
BM1B @ 6″	BM1	maple	38 7/8″	¾″ ×3	2¼″	9/16″	3 3/16″
↳ margin crosscuts	remainder of BM1B	maple	37⅜″	BM1B1 = 17¾″; BM1B2 = 19⅜″	37⅛″	¼″	1½″
BM1B1 @ 3 3/16″	remainder of BM1B	maple	17¾″	¾″ ×1 · ¼″ ×2	1¼″	9/16″	1⅜″
BM1B2 @ 3 3/16″	remainder of BM1B	maple	19⅜″	¼″ ×10	2½″	1⅞″	-1 3/16″
BC1A @ 7 3/16″	BC1	cherry	38 7/8″	1½″ ×3 · ¾″ ×2	6″	15/16″	¼″
BC2A @ 6½″	BC2	cherry	38 7/8″	1½″ ×1 · ¾″ ×5	5¼″	1⅛″	⅛″
GH-C @ 15/16″	*None*	cherry	38 7/8″	¾″ ×1	¾″	3/16″	0″
BW1A @ 6¼″	BW1	walnut	38 7/8″	3″ ×1 · ¾″ ×3	5¼″	¾″	¼″
BW2A @ 6 1/16″	BW2	walnut	38 7/8″	¾″ ×5	3¾″	15/16″	1⅜″

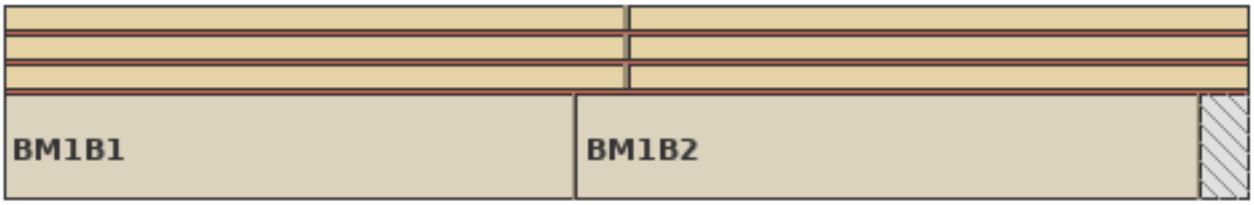
Board Rip Diagram

Review the board rip diagrams against the rip list.

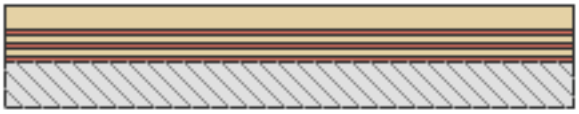
BM1A (38⅞″ x 6″ x 13⁄16″)



BM1B (38⅞″ x 6″ x 13⁄16″)



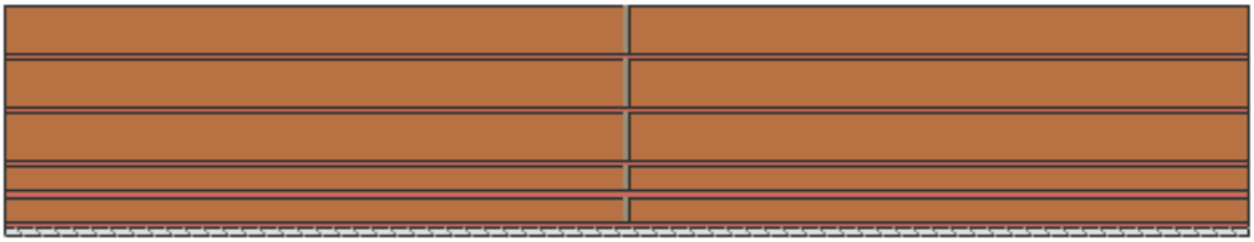
BM1B1 (17¾″ x 3 3⁄16″ x 13⁄16″)



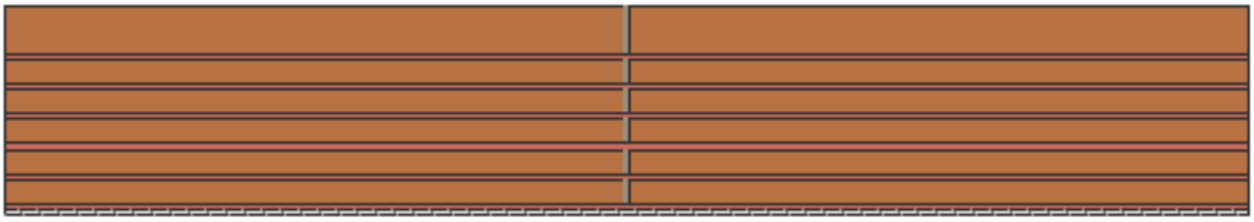
BM1B2 (19⅜″ x 3 3⁄16″ x 13⁄16″)



BC1A (38⅞″ x 7 3⁄16″ x 13⁄16″)



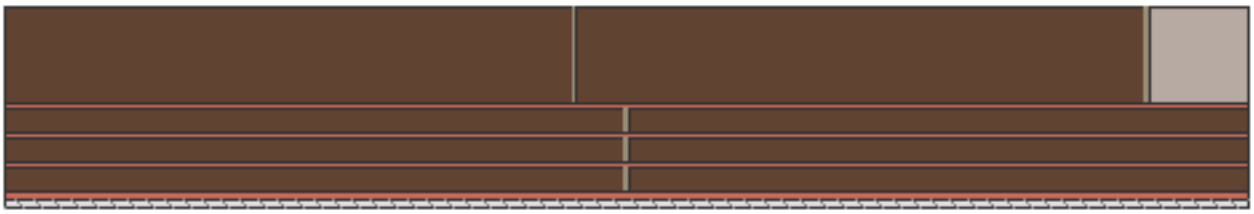
BC2A (38⅞″ x 6½″ x 13⁄16″)



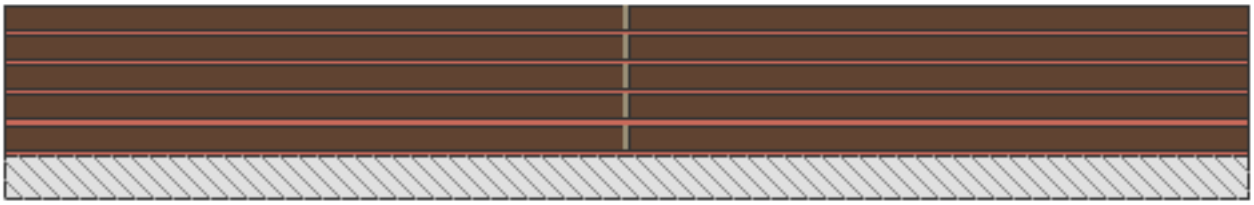
GH-C (Not Available)



BW1A (38⅞″ x 6¼″ x 13⁄16″)



BW2A (38⅞″ x 6 1⁄16″ x 13⁄16″)

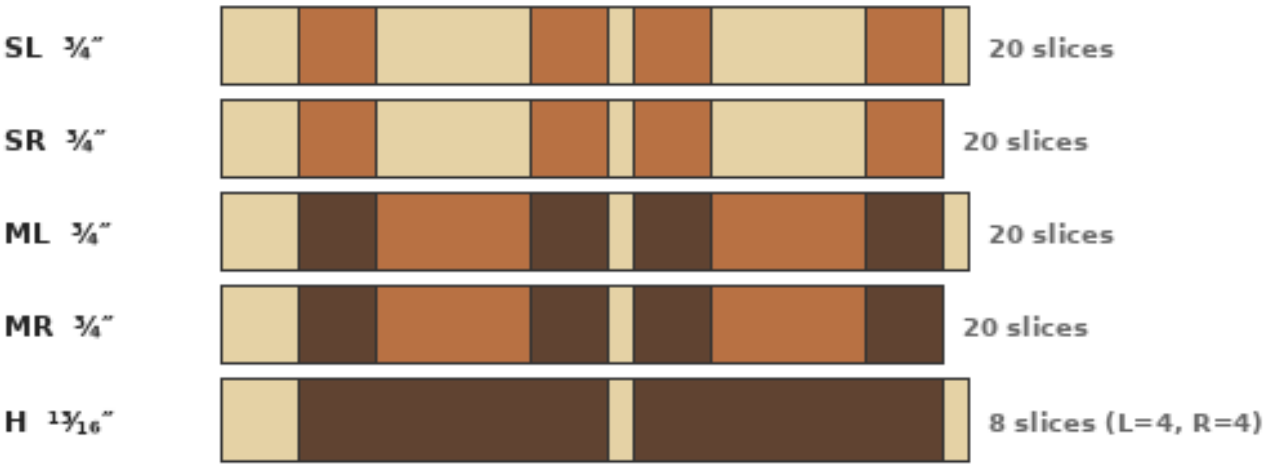


6. Panels overview

Panel	Qty	Length	Slices	Height	Ripped	Sequence (outer → in)
SL	2	19⅜″	11	¾″	—	Me · C2 · M1 · C2 · Md · C2 · M1 · C2 · Md
SR	2	19⅜″	11	¾″	—	Me · C2 · M1 · C2 · Md · C2 · M1 · C2
ML	2	19⅜″	11	¾″	—	Me · W2 · C1 · W2 · Md · W2 · C1 · W2 · Md
MR	2	19⅜″	11	¾″	—	Me · W2 · C1 · W2 · Md · W2 · C1 · W2
H	1	17¾″	10 (L:5 + R:5)	¼″	2× (¼″)	Me · W1 · Md · W1 · Md

Panel End Grain Diagram

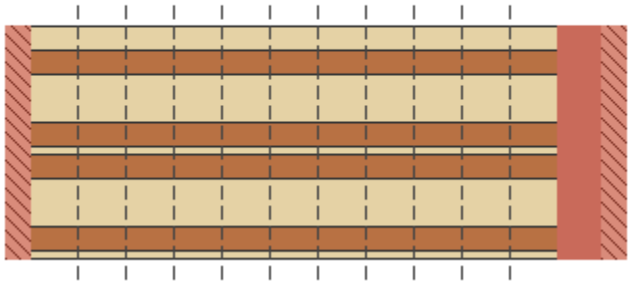
Review the panel end-grain diagrams.



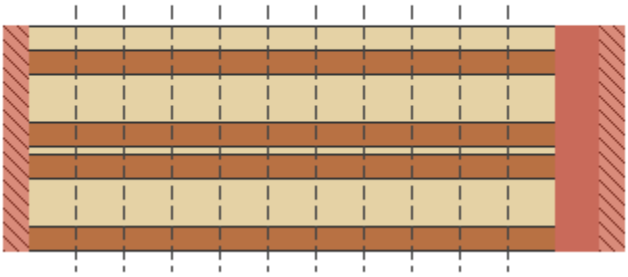
Panel Face Diagram

Review the panel face diagrams.

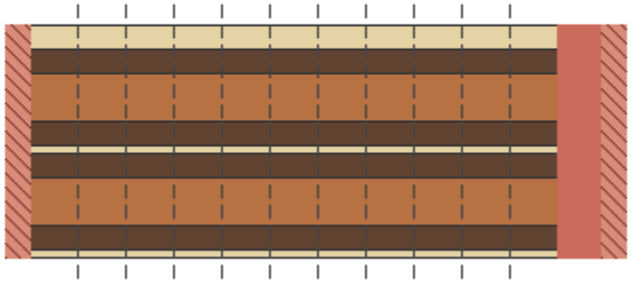
Panel SL ×2
19⅜″ long (11 x 2 = 22 slices)



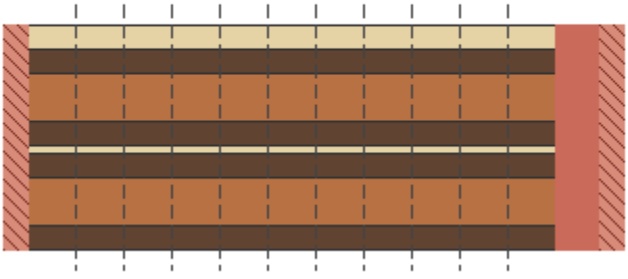
Panel SR ×2
19⅜″ long (11 x 2 = 22 slices)

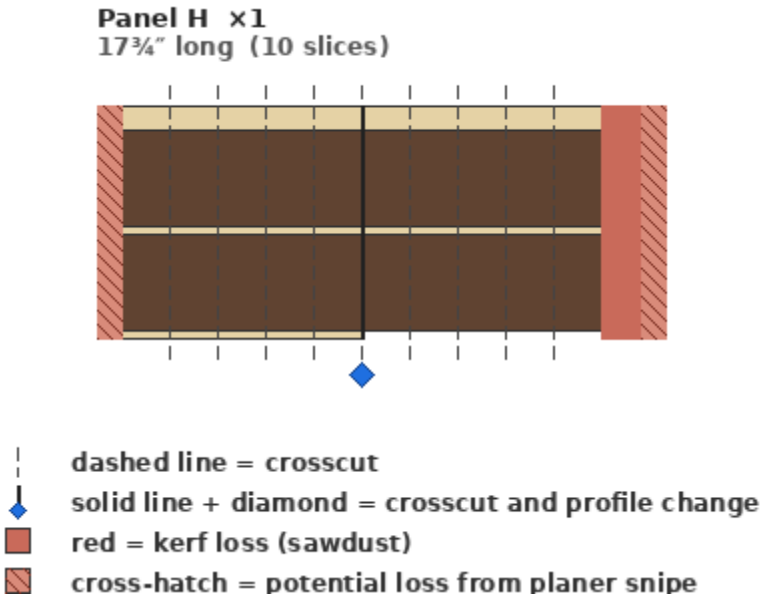


Panel ML ×2
19⅜″ long (11 x 2 = 22 slices)



Panel MR ×2
19⅜″ long (11 x 2 = 22 slices)





Slice Fulfillment

Review slice demand, supply, and offset.

Slice	Demand	Supplied	Supply Offset
SL	10 × 2 = 20	22	+2
SR	10 × 2 = 20	22	+2
ML	10 × 2 = 20	22	+2
MR	10 × 2 = 20	22	+2
H-L	4 × 2 = 8	5 × 2 = 10	+2
H-R	4 × 2 = 8	5 × 2 = 10	+2

7. Components

Review component demand, fulfillment, and shortfall.

Component	Species	Width	Length	Demand	Fulfilled	Shortfall
C1	cherry	1½″	19⅜″	8	8	0
C2	cherry	¾″	19⅜″	16	16	0
M1	maple	1½″	19⅜″	8	8	0
Md	maple	¼″	19⅜″	12	12	0
Md	maple	¼″	17¾″	2	2	0
Me	maple	¾″	19⅜″	8	8	0
Me	maple	¾″	17¾″	1	1	0
W1	walnut	3″	17¾″	2	2	0
W2	walnut	¾″	19⅜″	16	16	0

8. Panels (profiles and sequence)

Review the panels and their quantities.

Panel	Qty	Length	Profiles (slices)	Component sequence
SL	2	19⅜″	single: 11	Me · C2 · M1 · C2 · Md · C2 · M1 · C2 · Md
SR	2	19⅜″	single: 11	Me · C2 · M1 · C2 · Md · C2 · M1 · C2
ML	2	19⅜″	single: 11	Me · W2 · C1 · W2 · Md · W2 · C1 · W2 · Md
MR	2	19⅜″	single: 11	Me · W2 · C1 · W2 · Md · W2 · C1 · W2
H	1	17¾″	L: 5, R: 5	Me · W1 · Md · W1 · Md